

## Data Flow Diagram

|                      |                                                 |
|----------------------|-------------------------------------------------|
| <b>Date</b>          | 2 July 2026                                     |
| <b>Team ID</b>       | XXXXXX                                          |
| <b>Project Name</b>  | OptiCrop – Smart Agricultural Production Engine |
| <b>Maximum Marks</b> | 2 Marks                                         |

### Data Flow Diagram

#### DFD Symbol Legend

| Symbol                                   | Name                   | Description                                                                                                                                                    |
|------------------------------------------|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Oval / Rounded shape</b>              | <b>External Entity</b> | <b>Farmer</b> – The user who enters soil nutrient and weather details and receives the recommended crop.                                                       |
| <b>Rectangle with numbered header</b>    | <b>Process</b>         | <b>1. Crop Prediction Process</b> – Accepts user input, validates the values, sends them to the Machine Learning model, and generates the crop recommendation. |
| <b>Rectangle (solid fill, no number)</b> | <b>Data Store</b>      | <b>Crop Dataset (CSV) and Trained ML Model (model.pkl)</b> – Stores the agricultural dataset and the trained machine learning model used for prediction.       |
| <b>Labeled arrow</b>                     | <b>Data Flow</b>       | Represents the flow of data such as <b>Soil &amp; Weather Inputs, Prediction Request, and Recommended Crop</b> between the farmer, process, and data stores.   |

### Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

